Research Monograph

From Sensor Signatures to Survival Curves

A Physics-Regularized Digital Twin for Milling Tool Wear and Tool-Life Forecasting

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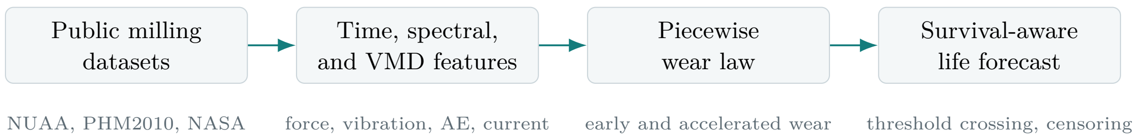

Local workspace: `sop-project`

Compiled from Phase 2, lifetime extraction, cross-dataset, VMD, and interpretability artifacts

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Abstract

This monograph consolidates the tool-wear research contained in the local `sop-project` repository into a clean, reproducible research narrative. The work studies milling tool wear, tool-health failure detection, and threshold-crossing tool-life estimation using three complementary public data sources: NUAA/UniWear orthogonal milling experiments, PHM2010 cutter wear traces, and the NASA milling run-to-failure dataset. The research objective is not only to reduce pointwise wear error, but to build an interpretable and analytically useful model that can drive a simulator, explain load effects, support uncertainty bands, and connect signal-processing evidence to practical tool replacement decisions.

The central result is a physics-regularized piecewise wear law. The early regime is a sub-linear power law with exponent $m_{\mathrm{e}} = 0.64$, verified on NUAA and PHM2010. The late regime is an accelerated power law with exponent $m_{\mathrm{l}} = 1.29$, extracted from NASA run-to-failure behavior. Around the reference cutting condition $(n, f_z, a_p) = (1800\ \mathrm{rpm}, 0.05\ \mathrm{mm/tooth}, 3.0\ \mathrm{mm})$, the early amplitude is governed by

$$\phi = \left(\frac{n}{1800}\right)^{0.359967} \left(\frac{f_z}{0.05}\right)^{3.956013} \left(\frac{a_p}{3.0}\right)^{0.688073},$$

with feed emerging as the strongest load driver. The research then extends this equation into a lifetime-modeling study. Wear models were trained on NUAA, inverted into threshold-crossing life estimates, and compared with censored accelerated-failure-time equations. The best pointwise wear model, Gradient Boosting, did not produce the best event-life forecasts after inversion. This negative result is central: optimizing wear RMSE is not equivalent to optimizing threshold-crossing life.

The report also records frequency-domain evidence from variational mode decomposition (VMD), feature-selection stability, SHAP interpretation, cross-dataset regression and classification metrics, residual diagnostics, bootstrap coefficient uncertainty, life prediction intervals, and early-window calibration tests. The final system should be interpreted as a research-grade digital twin prototype: it is useful for explanation, scenario ranking, and simulator behavior, but still needs broader ingestion of newer 2025 full-life milling datasets before it should be treated as an industrial replacement policy.

Chapter 1

Research Question and Contributions

1.1 Problem

Milling tool wear is not a single regression problem. A tool can spend long periods in a slow early-wear regime, then enter an accelerated damage regime where a small amount of additional cutting time produces a large increase in flank wear. Datasets emphasize different parts of this story. Some datasets capture many process settings but short or synthetic lives. Others contain full run-to-failure trajectories but few cutting conditions. A useful research model therefore has to answer four questions at once:

1. What wear curve explains the physical progression from early wear to accelerated wear?
2. How do spindle speed, feed, and depth alter the rate of that curve?
3. Which sensor signatures provide evidence that the tool is moving toward a failure threshold?
4. Can a model trained on wear observations be inverted into a practical lifetime equation?

The local repository addresses these questions with a staged approach. The first stage extracts signal features and cross-dataset baselines. The second stage builds a piecewise wear equation. The third stage extends the equation with uncertainty and residual diagnostics. The fourth stage explicitly trains wear models, inverts them into lifetime predictions, and compares them with survival-style accelerated-failure-time models.

1.2 Main Contributions

- A dataset-role framework that uses NUAA for load-dependent early wear, PHM2010 for early-exponent transfer, and NASA for accelerated late-stage behavior.
- A compact piecewise wear model with interpretable exponents and a process-load intensity function.
- A cross-dataset health pipeline that evaluates wear regression and failure classification under grouped validation.
- VMD-based frequency evidence showing that decomposed vibration, acoustic-emission, and spindle-current signatures improve the strongest absolute correlation with wear from 0.4806 to 0.5607.
- A lifetime-model extraction pipeline that trains multiple wear models, inverts them into threshold-crossing estimates, and exposes the mismatch between wear RMSE and life accuracy.
- Censored accelerated-failure-time equations for NUAA and material-aware NASA life behavior.
- Bootstrap uncertainty, residual diagnostics, and calibration tests that describe where the current model is stable and where it remains under-identified.

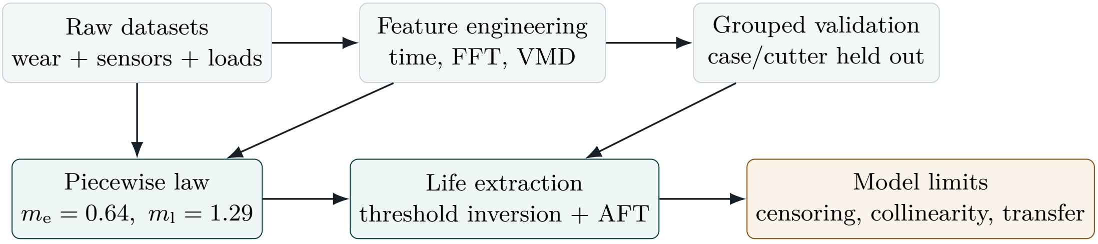


Figure 1.1: Research architecture followed in the repository. The workflow separates signal evidence, physics-regularized equation building, grouped validation, and survival-aware life estimation.

Chapter 2

Data Sources and Evidence Roles

2.1 Dataset Inventory

The repository contains or references three primary milling sources. They are not interchangeable, so the research assigns each dataset a specific inferential role. This is preferable to pooling everything into one opaque regression table, because the datasets differ in time base, sensors, process variables, material, failure threshold, and whether the experiment reaches severe wear.

Table 2.1: Primary datasets and their research roles.

Dataset	Local role	Main measurements	Why it matters
NUAA / Uni-Wear	Early wear and process-condition law	Wear, spindle speed, feed per tooth, axial depth	Supplies process variation needed to fit a load-intensity function and lifetime surfaces. It includes censored runs, which makes survival-aware modeling necessary.
PHM2010	Early-regime transfer check	Cutter wear traces and derived features	Tests whether the early exponent learned from NUAA is a local artifact or a transferable early-wear slope.
NASA Milling	Late-stage acceleration and material behavior	Vibration, acoustic emission, spindle current, wear, feed, depth, material	Contains severe wear progression and rich sensors, making it the best local source for accelerated wear, frequency-domain evidence, and material-aware survival behavior.

2.2 Data Role Diagram

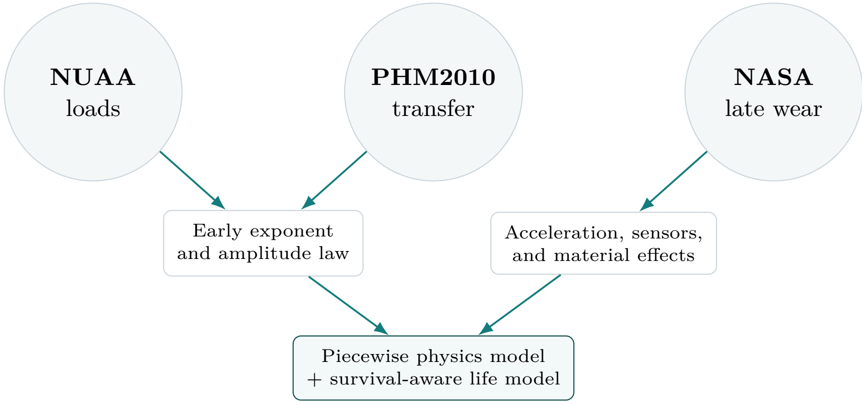


Figure 2.1: The datasets act like complementary evidence sources. NUAA and PHM2010 constrain the early regime, while NASA constrains late acceleration and sensor signatures.

2.3 Failure Thresholds

The failure classification task uses dataset-specific thresholds because the measurement scale and experimental intent differ. The thresholds are not presented as universal industrial replacement limits; they are operational labels used to compare health markers within each dataset.

Table 2.2: Dataset-specific tool-health failure labels used in the cross-dataset research pipeline.

Dataset	Samples	Threshold	Failure rate
PHM2010	945	0.150 mm	0.1725
NASA	146	0.300 mm	0.4795
UniWear	649	0.220 mm	0.2743

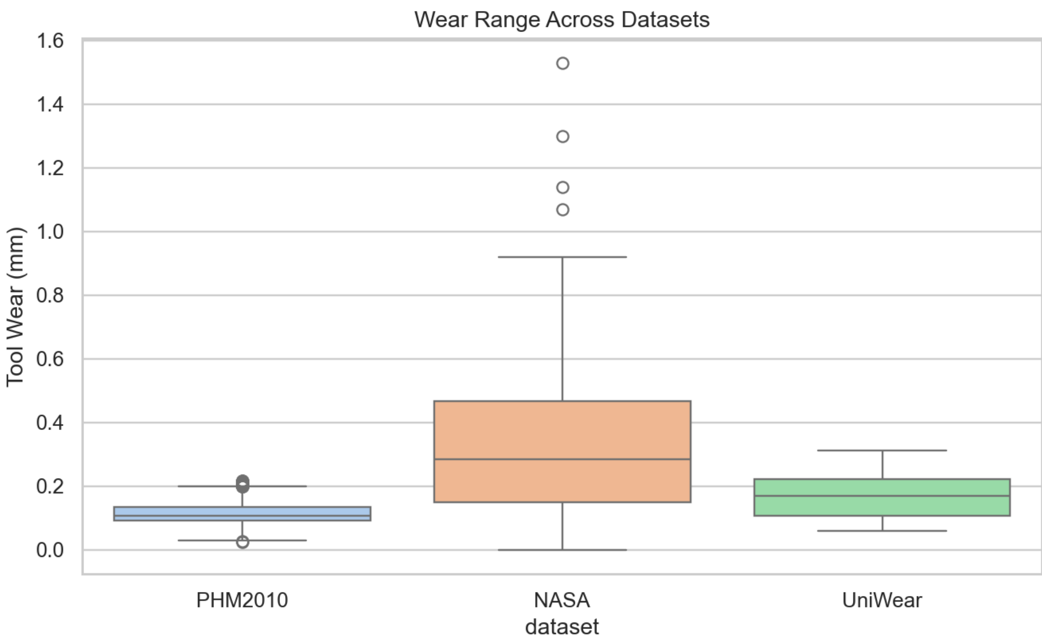


Figure 2.2: Wear distributions across local datasets. The different ranges motivate dataset-specific thresholds and role-specific modeling rather than direct pooling.

Chapter 3

Feature Engineering and Frequency Evidence

3.1 Feature Families

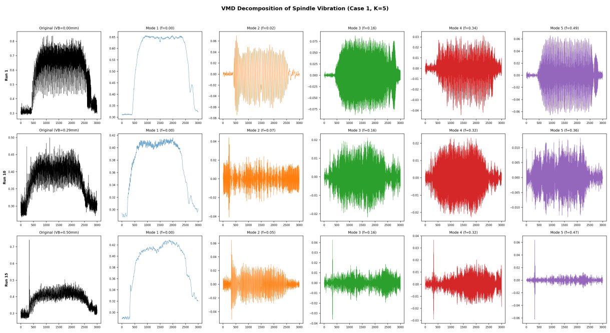
The repository extracts compact signal features from time-domain and frequency-domain measurements. The shared feature vocabulary includes mean, standard deviation, root-mean-square, absolute mean, peak-to-peak range, skew-like statistics, kurtosis-like statistics, crest factor, quantiles, FFT energy, dominant FFT bin, FFT centroid, and spectral entropy. The NASA pipeline also includes VMD features computed over vibration, acoustic-emission, spindle-current AC, and spindle-current DC channels.

Table 3.1: Feature families and interpretation.

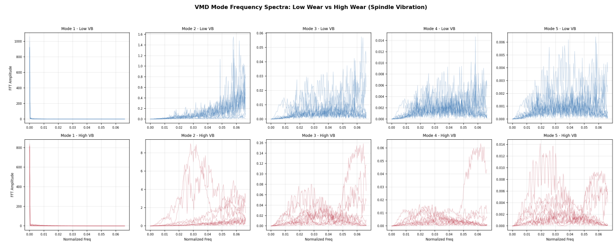
Family	Examples	Tool-wear interpretation
Time-domain level	mean, RMS, absolute mean	Captures load and energy growth as cutting becomes less efficient.
Time-domain shape	skewness, kurtosis, crest factor, quantiles	Captures impulsive events, chatter bursts, rubbing, and intermittent edge damage.
Frequency-domain energy	FFT energy, FFT peak, centroid, entropy	Captures shifts in vibration and acoustic-emission structure as wear changes cutting dynamics.
VMD mode features	mode energy, entropy, center frequency, mode kurtosis	Separates mixed sensor signals into interpretable bands, improving some wear correlations over raw features.
Process variables	speed, feed, axial depth, material	Anchor the statistical model in machining physics and enable scenario simulation.

3.2 Variational Mode Decomposition

The VMD pass decomposes NASA signals into $K = 5$ modes with $\alpha = 2000$ and $\tau = 0$. It produces 390 VMD features over sensors `vib_spindle`, `vib_table`, `AE_table`, `AE_spindle`, `smcAC`, and `smcDC`. The best raw feature correlation with wear was $|r| = 0.4806$. The best VMD feature reached $|r| = 0.5607$, an improvement of 16.7%.

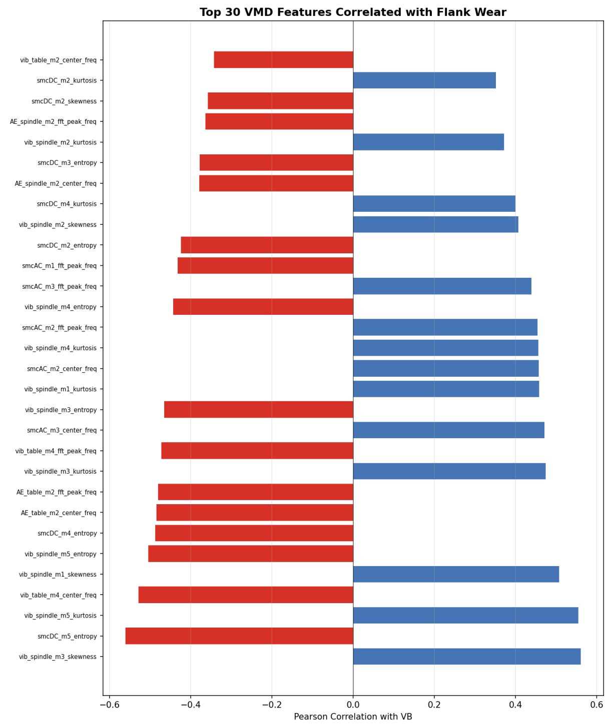


(a) Example decomposition

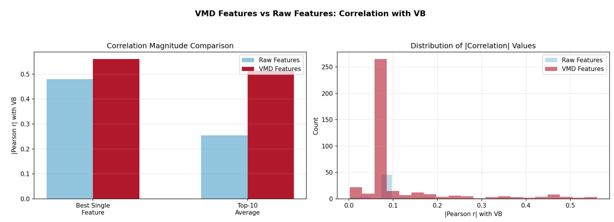


(b) Mode spectra

Figure 3.1: VMD separates broadband machining signals into mode-specific frequency content before feature extraction.



(a) Top VMD correlations



(b) Raw versus VMD comparison

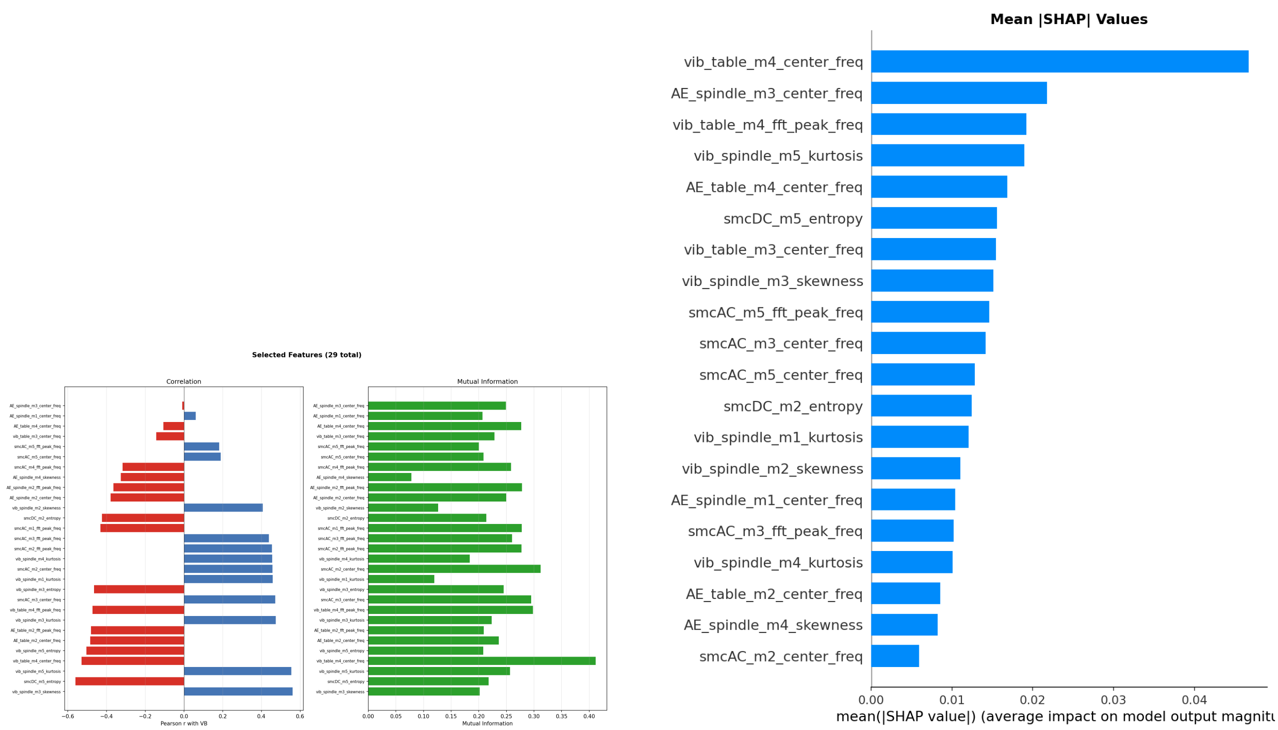
Figure 3.2: Frequency-mode features provide stronger individual wear associations than the best raw features in this NASA analysis.

Table 3.2: Top VMD features by absolute correlation with NASA flank wear.

Feature	r	$\|r\|$	Interpretation
`vib_spindle_m3_skewness`	0.5607	0.5607	Wear increases asymmetric spindle vibration bursts.
`smcDC_m5_entropy`	-0.5604	0.5604	Current structure becomes less evenly distributed in the high-order mode.
`vib_spindle_m5_kurtosis`	0.5542	0.5542	Impulsive high-frequency vibration grows with tool damage.
`vib_table_m4_center_freq`	-0.5287	0.5287	Table vibration frequency content shifts as cutting stiffness changes.
`vib_spindle_m1_skewness`	0.5079	0.5079	Low-mode spindle asymmetry tracks progressive wear.

3.3 Feature Selection and Interpretation

Feature selection and SHAP analysis support the same broad conclusion: the useful predictors are not only raw amplitude measures. Wear information appears in mode frequency, entropy, kurtosis, and acoustic-emission center-frequency features.



(a) Selected feature ranking (b) SHAP importance

Figure 3.3: Selected and interpreted features emphasize VMD-derived vibration, acoustic-emission, and spindle-current information.

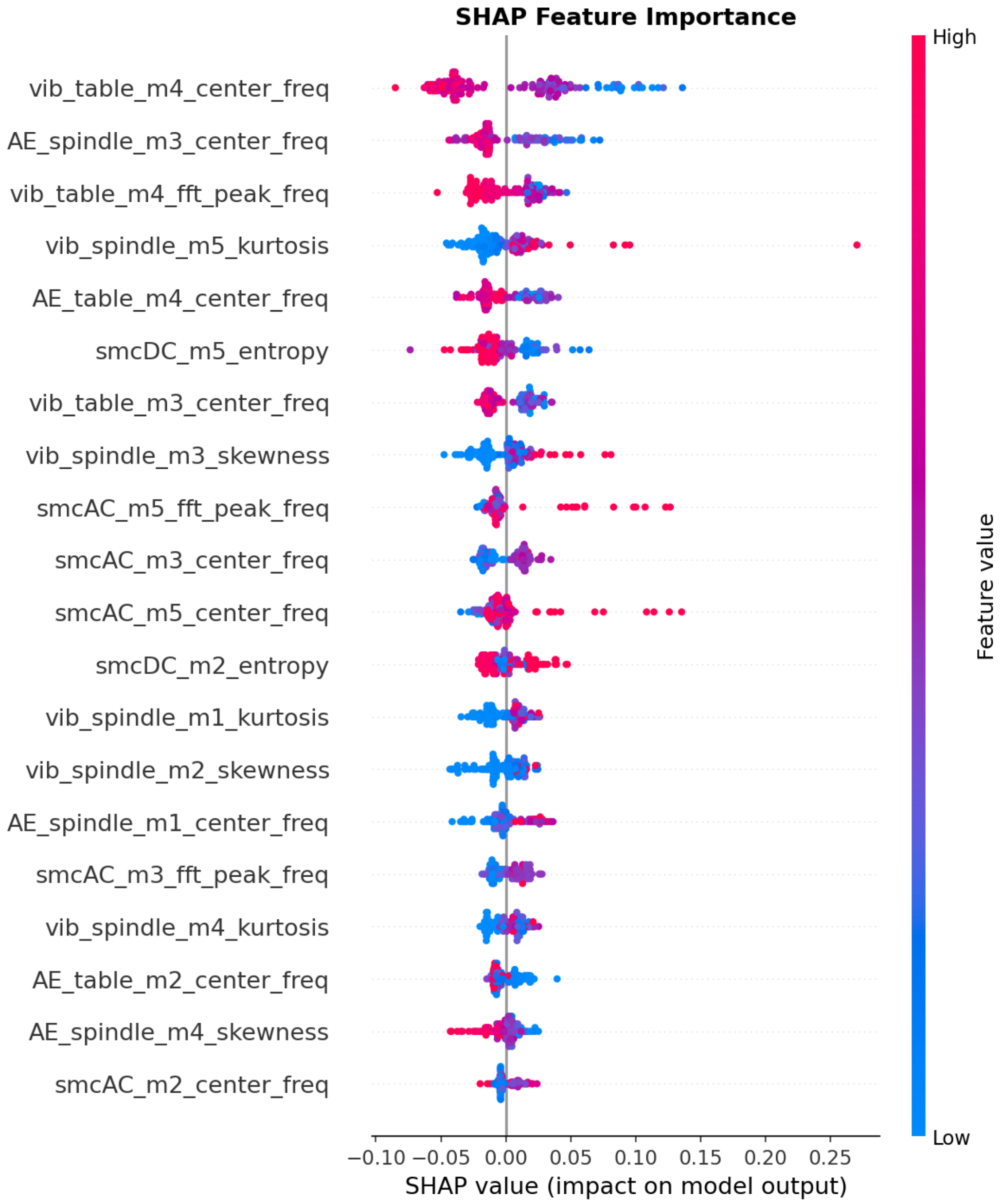


Figure 3.4: SHAP summary plot for the NASA modeling pass. High-impact features include VMD center-frequency, entropy, and kurtosis terms.

Chapter 4

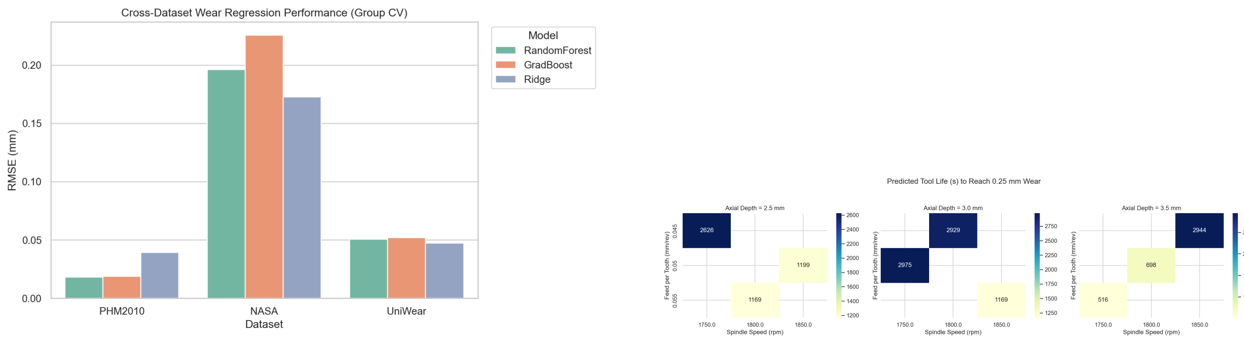
Cross-Dataset Baselines

4.1 Regression and Failure Classification

The cross-dataset pipeline evaluates Ridge, Random Forest, and Gradient Boosting style models with grouped validation. Grouped validation is important because adjacent samples from the same cutter or case are not independent deployment tests. The goal is to estimate whether a model transfers to unseen experiments, not whether it can interpolate adjacent points within the same trajectory.

Table 4.1: Cross-dataset wear regression and failure classification results.

Dataset	Samples	Groups	Features	Best regressor	RMSE	ROC-AUC
PHM2010	945	3	105	Random Forest	0.0181	0.9671
NASA	146	16	94	Ridge	0.1728	0.9141
UniWear	649	12	46	Ridge	0.0475	0.7350



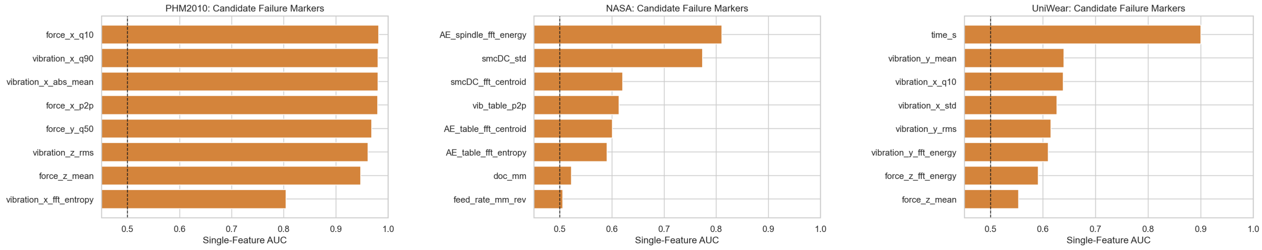
(a) Regression RMSE

(b) NUAA life heatmap

Figure 4.1: Cross-dataset baselines and NUAA process-condition sensitivity.

4.2 Per-Dataset Health Markers

The failure-marker plots show a practical theme: failure labels are usually preceded by increasing signal energy, dispersion, or impulsiveness, but the exact feature family varies by dataset. This is consistent with machining physics. Tool wear changes the chip formation process, contact geometry, friction, heat generation, and dynamic stiffness; the resulting evidence can appear in vibration, acoustic emission, force/current, or derived frequency features depending on available sensors.



(a) PHM2010 (b) NASA (c) UniWear

Figure 4.2: Failure-marker summaries across datasets. The plots support the use of energy, dispersion, crest, and modal frequency indicators as health signals.

Chapter 5

Piecewise Physics-Regularized Wear Law

5.1 Why a Piecewise Law

The repository's Phase 2 model uses a piecewise curve rather than a single global polynomial or black-box regressor. The reason is physical and statistical. Early wear often grows sub-linearly after initial edge conditioning. Late wear accelerates when thermal, mechanical, and geometric effects amplify each other. A single smooth exponent must compromise between these regimes; the compromise can fit average wear but misrepresent lifetime near the failure threshold.

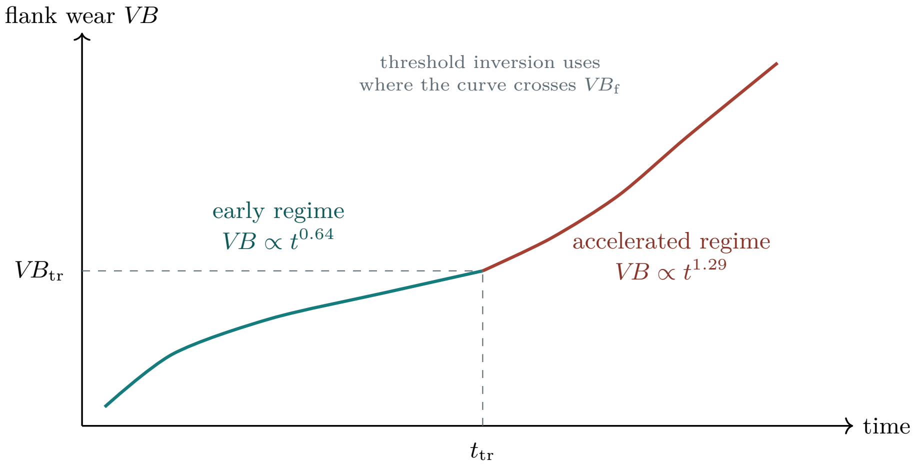


Figure 5.1: Schematic of the piecewise wear law. The transition wear is $VB_\mathrm{tr} = 0.25$ mm; early and late exponents are learned from different evidence sources.

5.2 Model Definition

Let t denote cutting time in minutes for the simulator-scale equation, and let VB_f denote the target failure threshold. The Phase 2 model defines an early amplitude law around a reference operating point:

$$\phi(n, f_z, a_p) = \left(\frac{n}{1800}\right)^{0.359967} \left(\frac{f_z}{0.05}\right)^{3.956013} \left(\frac{a_p}{3.0}\right)^{0.688073}. \tag{5.1}$$

The early wear curve is

$$VB(t) = k_\mathrm{early}\ \phi\ t^{m_\mathrm{e}}, \quad k_\mathrm{early} = 0.018580943, \quad m_\mathrm{e} = 0.64. \tag{5.2}$$

The late curve is activated after the transition wear $VB_{\mathrm{tr}} = 0.25$ mm. It uses an accelerated exponent $m_{\mathrm{l}} = 1.29$ and material-aware late-stage factors. In simulator form, the model can be summarized as

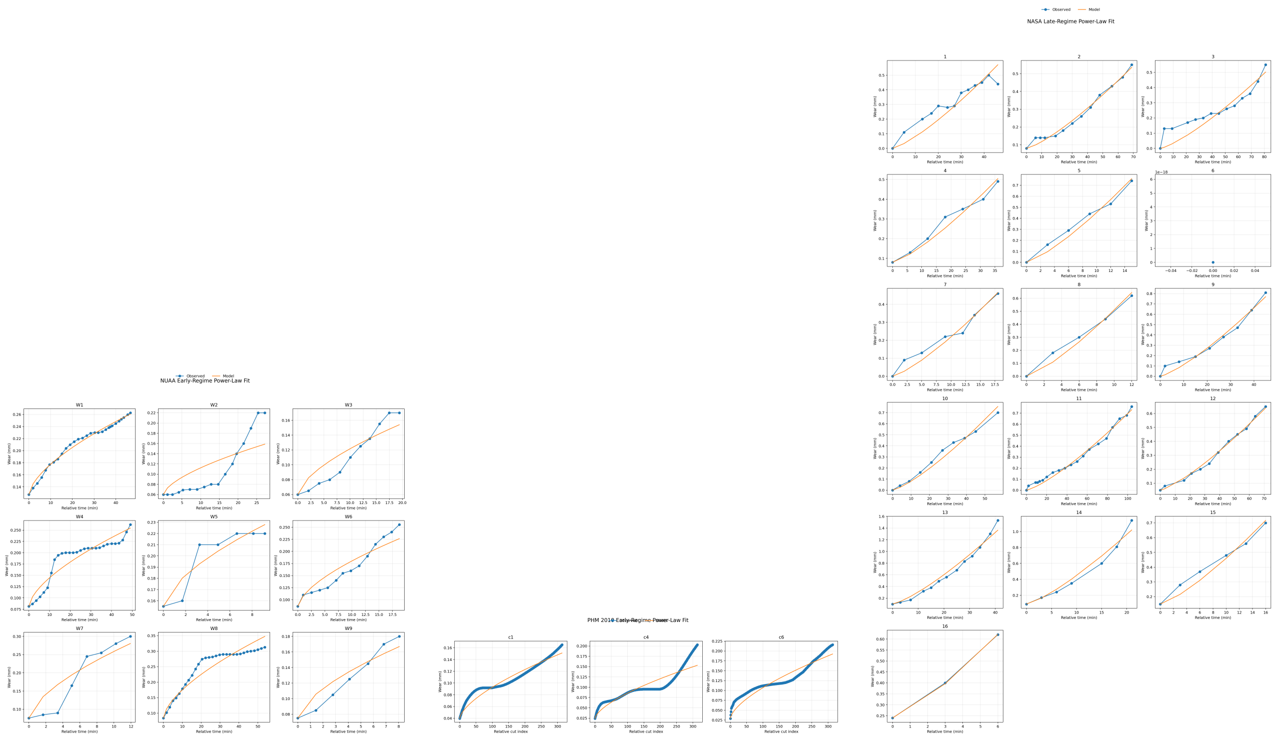
$$VB(t) = \begin{cases} k_{\mathrm{early}}\ \phi\ t^{0.64}, & VB(t) < VB_{\mathrm{tr}}, \\ VB_{\mathrm{tr}} + k_{\mathrm{late}}(m)\ \phi\ (t - t_{\mathrm{tr}})^{1.29}, & VB(t) \geq VB_{\mathrm{tr}}, \end{cases} \tag{5.3}$$

where $k_{\mathrm{late}}(m)$ is adjusted by material family. The repository records the late-stage factors as generic 0.25, cast iron 0.1488, and steel 0.3994. These factors should be interpreted as local calibrated simulator coefficients, not universal constants.

5.3 Evidence for Exponents

Table 5.1: Evidence used to select the Phase 2 exponents.

Dataset	Regime	Raw exponent	RMSE	Role in final model
NUAA	Early	0.6347	0.0211 mm	Main source for the sub-linear early wear law.
PHM2010	Early	0.6471	0.0136 mm	Independent confirmation that the early exponent transfers.
NASA	Late	1.2873	0.0438 mm	Main source for accelerated late-stage behavior.

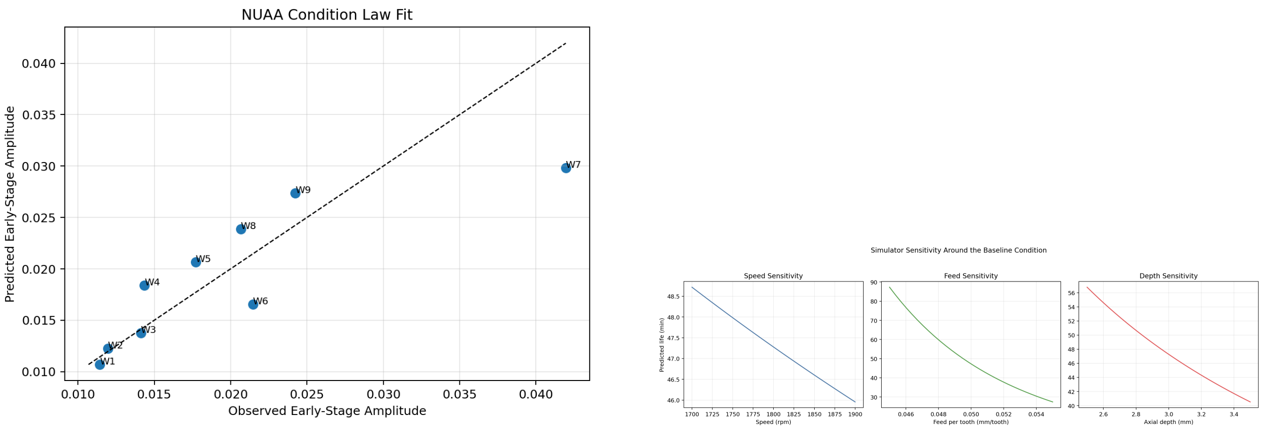


(a) NUAA early fit (b) PHM2010 check (c) NASA late fit

Figure 5.2: The final exponents are grounded in role-specific fits rather than a single pooled curve.

5.4 Condition Law

The fitted process-condition intensity in Equation (5.1) is strongest in feed. That result is plausible: feed directly changes chip thickness and cutting load per tooth. Speed and depth also matter, but in the local NUAA design the feed exponent is much more stable and much larger than the others.



(a) Condition-law fit

(b) Life sensitivity curves

Figure 5.3: Feed dominates the local life sensitivity, while speed and depth remain less stable because of narrower design variation and collinearity.

5.5 Material Effects

NASA supports late-stage material comparisons, especially between cast iron and steel. The Phase 2 simulator uses this evidence to avoid treating all late-stage wear as one material-independent acceleration process.

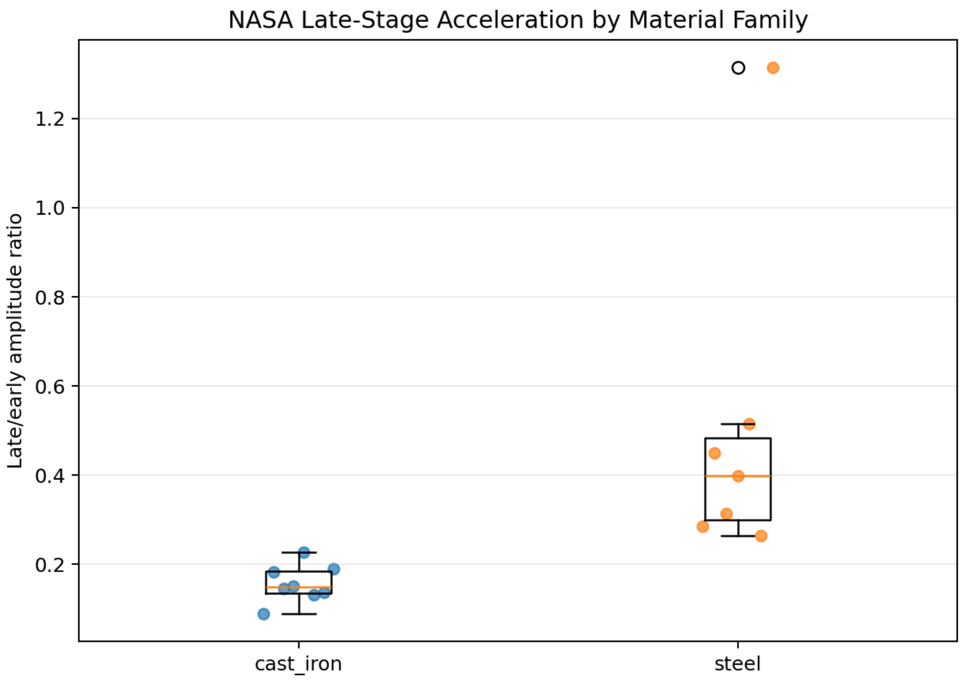


Figure 5.4: NASA late-stage material behavior. Steel shows a stronger late-stage wear factor than cast iron in the local model.

Chapter 6

Uncertainty, Calibration, and Residuals

6.1 Bootstrap Coefficient Stability

The extended research pass bootstraps the NUAA condition law with valid-design filtering. This is important because small experimental designs can produce rank-deficient or unstable resamples. The bootstrap results show a stable feed exponent, while speed and depth remain more uncertain.

Table 6.1: Bootstrap coefficient summary for the NUAA condition law.

Quantity	2.5%	Median	97.5%
k_{early}	0.01558	0.01783	0.02183
Speed exponent	-7.49808	1.16739	8.04424
Feed exponent	2.04563	3.99243	6.68021
Depth exponent	-0.59012	0.51957	1.92473
Amplitude R^2	–	0.89715	–

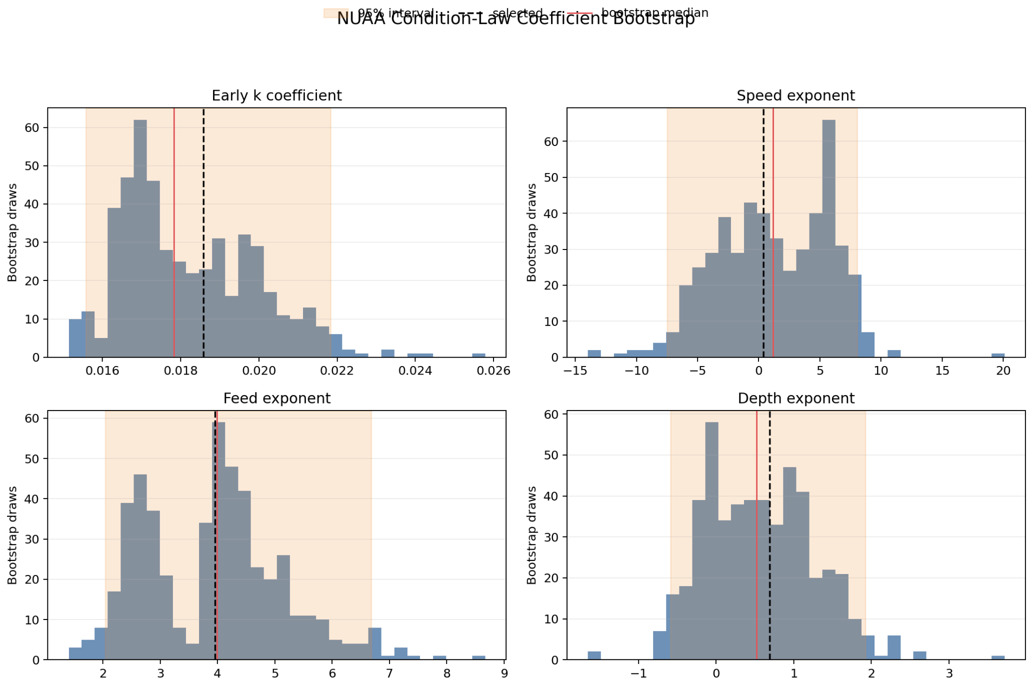


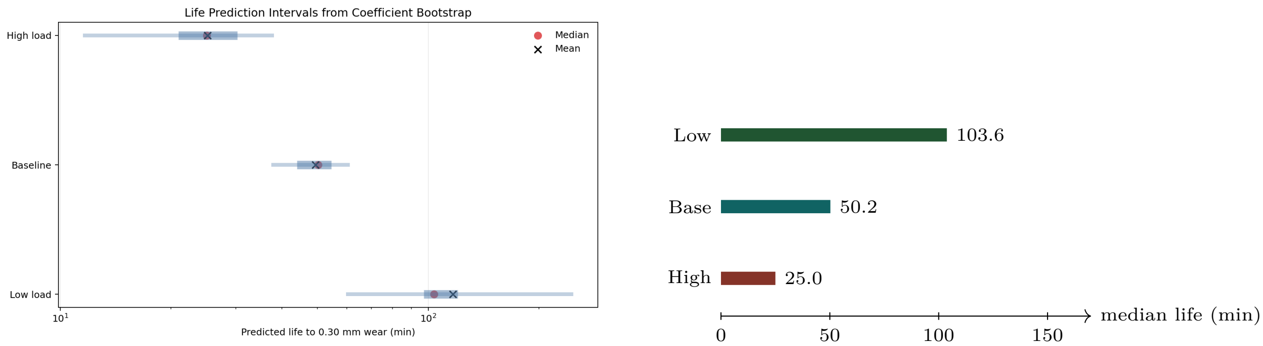
Figure 6.1: Bootstrap distributions for condition-law coefficients. Feed is the most consistently identified operating variable in the local design.

6.2 Life Prediction Intervals

The bootstrap draws were propagated into low-load, baseline, and high-load life scenarios. The result gives a practical load comparison: the baseline median life is about 50.2 minutes, the low-load condition extends median life to about 103.6 minutes, and the high-load condition compresses median life to about 25.0 minutes.

Table 6.2: Bootstrap life prediction intervals.

Scenario	Speed	Feed	Depth	2.5%	Median	97.5%
Low load	1750	0.045	2.5	59.797	103.644	247.675
Baseline	1800	0.050	3.0	37.422	50.227	61.177
High load	1850	0.055	3.5	11.514	24.987	38.046



(a) Interval estimates (b) Load comparison sketch

Figure 6.2: Load changes produce large predicted life changes; the high-load condition roughly halves baseline life in the bootstrap median.

6.3 Early Calibration Result

The extended research pass tested whether early observations from each run could calibrate the global equation through a per-run multiplier. This is a reasonable idea: if the first few points indicate that a tool is wearing faster than expected, a calibrated forecast might improve the remaining trajectory. The local NUAA holdout result was negative. For the tested early cutoffs, calibrated forecasts had worse median holdout RMSE than the uncalibrated global model.

Table 6.3: NUAA early-window calibration holdout. Lower RMSE is better.

Cutoff fraction	Experiments	Median points	Calibrated RMSE	Uncalibrated RMSE
0.25	7	5.0	0.05045	0.04047
0.40	8	4.5	0.04750	0.02508
0.60	8	7.0	0.04230	0.02819

This result should stay in the paper because it prevents an overly optimistic simulator story. The current global model is more stable than short-window multiplicative recalibration. In future work, calibration should use a structured Bayesian update, monotonic constraints, or a state-space model instead of a free scalar multiplier.

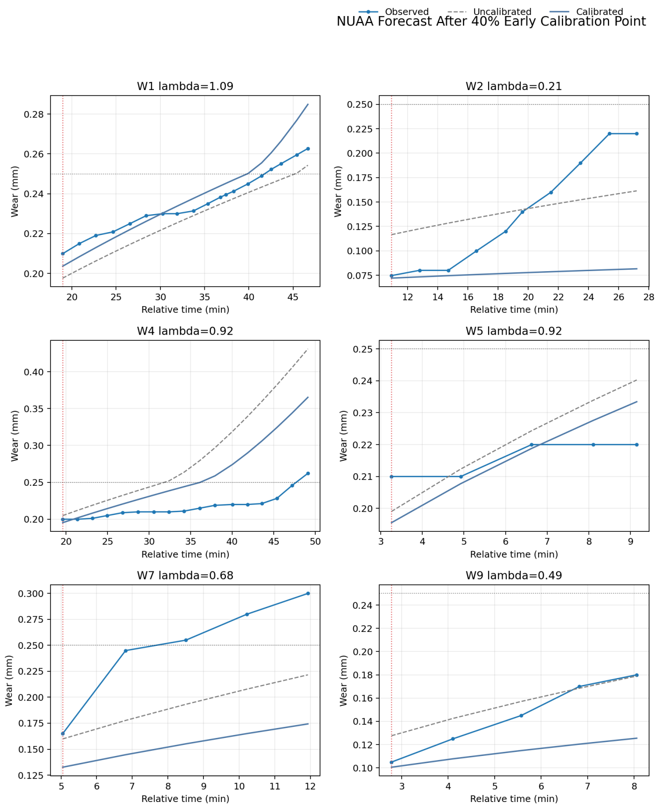


Figure 6.3: Calibration forecast diagnostic. Short-window calibration can overcorrect and worsen holdout forecasts in the local NUAA experiment set.

6.4 Residual Diagnostics

Residual diagnostics show that the piecewise equation is most accurate in PHM2010 and NUAA early regimes, and least accurate in NASA, where severe wear, material variation, and late-stage behavior are stronger.

Table 6.4: Residual diagnostics for the Phase 2 equation.

Dataset	Points	Bias	MAE	RMSE	95% abs. error
NASA	146	0.00763	0.03149	0.04376	0.08825
NUAA	152	-0.00162	0.01568	0.02113	0.04087
PHM2010	945	0.00104	0.01070	0.01362	0.02437

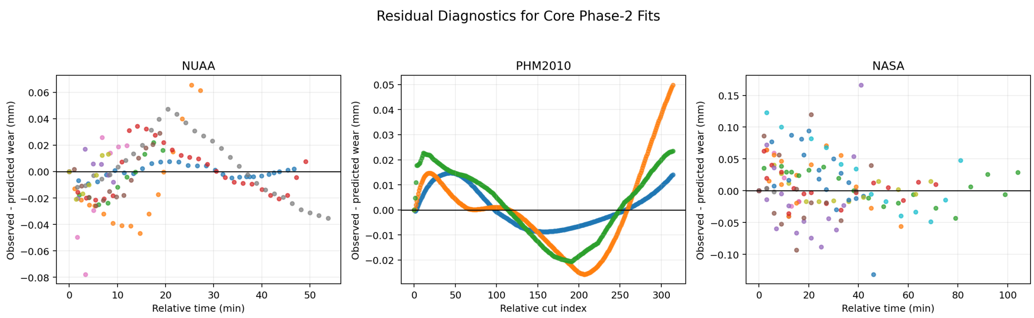


Figure 6.4: Residual diagnostics across datasets. The NASA tail remains the hardest local regime.

Chapter 7

Lifetime Modeling by Training and Inverting Models

7.1 Why Wear Accuracy Is Not Enough

The lifetime rework directly addressed whether tool life was estimated by training models and extracting results from those models. The answer is yes: NUAA wear models were trained, evaluated under leave-one-experiment-out validation, and inverted to estimate threshold-crossing life. The inversion step changes the target. A model can have low pointwise wear RMSE and still predict a poor crossing time if it is biased near the threshold.

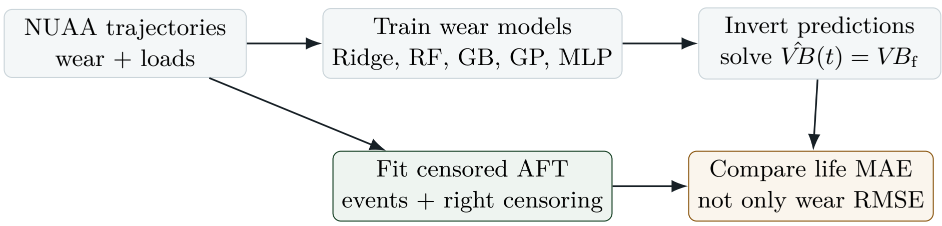


Figure 7.1: Lifetime extraction workflow. The repository trains wear models, inverts them into crossing times, and compares them with survival-style equations.

7.2 Threshold-Crossing Labels

Life labels were built by finding when each run crossed a dataset-specific wear threshold. Uncrossed runs were treated as censored. The time units were not mixed: NUAA and NASA lives are in minutes, while PHM2010 uses cutter cut index because its native time base differs.

Table 7.1: Threshold-crossing life labels.

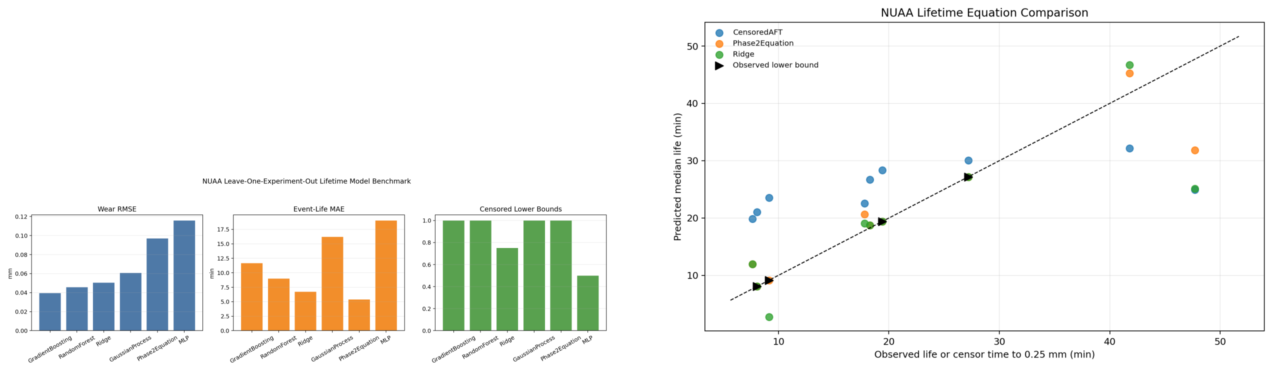
Dataset	Records	Threshold	Events	Censored
NUAA	9	0.250 mm	5	4
NASA	15	0.300 mm	15	0
PHM2010	3	0.150 mm	3	0

7.3 Trained Wear Models and Inverted Life

Table 7.2: NUAA trained wear models and inverted event-life performance.

Model	Wear RMSE	Wear MAE	Wear R^2	Event-life MAE	Censor bound rate
Phase2Equation	0.0969	0.0460	-0.8811	5.3832	1.00
Ridge	0.0505	0.0407	0.4883	6.7010	0.75
Random Forest	0.0457	0.0337	0.5817	8.9847	1.00
Gradient Boosting	0.0394	0.0310	0.6896	11.6555	1.00
Gaussian Process	0.0607	0.0459	0.2631	16.1923	1.00
MLP	0.1159	0.0956	-1.6890	19.0334	0.50

The best pointwise wear model is Gradient Boosting, but its event-life MAE is worse than the Phase 2 equation and Ridge. This is a strong negative result. It says the lifetime problem cannot be judged only by VB regression error. The selected model for extracting an interpretable life surface is Ridge because it is smoother and more reliable for inversion, even though it is not the lowest-RMSE wear model.



(a) Wear and life benchmark

(b) Life equation comparison

Figure 7.2: Wear-model inversion changes the model-selection problem. Smoothness near the threshold matters.

7.4 Censored AFT Equation

The NUAA censored accelerated-failure-time model is

$$T_{\min} = 24.93 \left(\frac{n}{1800}\right)^{-0.0187} \left(\frac{f_z}{0.05}\right)^{-1.7662} \left(\frac{a_p}{3.0}\right)^{-0.3768}, \tag{7.1}$$

with $\sigma = 0.5291$ and negative log likelihood 21.2365. The AFT feed exponent is weaker than the Phase 2 condition-law feed exponent because it models sparse threshold-crossing events with censoring, not dense early wear amplitudes.

Table 7.3: NUAA AFT bootstrap coefficients.

Coefficient	2.5%	Median	97.5%
Intercept	2.7199	3.2103	3.8286
log speed	-6.2471	-0.2186	0.9129
log feed	-7.4945	-2.1886	0.1153
log depth	-2.4450	-0.5467	4.5558
sigma	0.0000	0.1826	0.6821

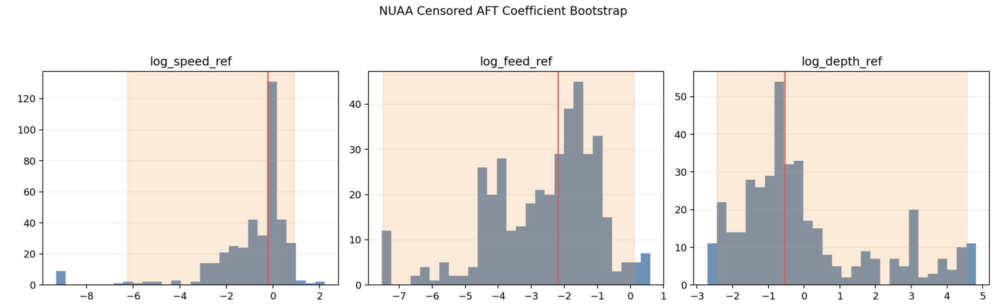


Figure 7.3: NUAA AFT bootstrap uncertainty. Sparse events and censoring make speed and depth effects weakly identified.

7.5 ML-Inverted Life Surface

The Ridge response surface was inverted over a grid of operating scenarios and then compressed into a Taylor-like equation:

$$T_{\min} = 47.52 \left(\frac{n}{1800}\right)^{-10.4924} \left(\frac{f_z}{0.05}\right)^{-4.6506} \left(\frac{a_p}{3.0}\right)^{-0.1372}, \tag{7.2}$$

with surrogate log-space $R^2 = 0.9800$ over 345 crossing scenarios. The high speed exponent should not be overinterpreted. It is a warning that the local speed range is narrow and speed is partly confounded with other load effects. The feed exponent is more consistent with the Phase 2 condition law and observed-life equation.

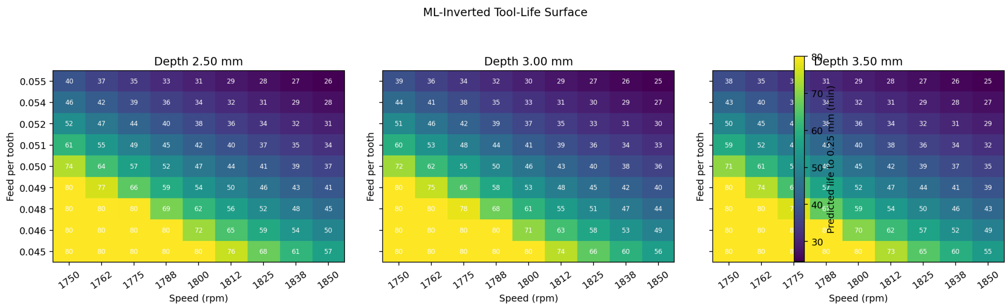


Figure 7.4: ML-inverted NUAA life surface. The surface is useful for scenario ranking but must be interpreted through the design limits of the training data.

7.6 Direct Observed-Life Equation

Using only uncensored NUAA threshold crossings yields

$$T_{\min} = 16.78 \left(\frac{n}{1800}\right)^{-12.6655} \left(\frac{f_z}{0.05}\right)^{-4.1634} \left(\frac{a_p}{3.0}\right)^{-2.4111}, \tag{7.3}$$

with log RMSE 0.3784. This equation is less robust because it discards censored runs. It is nevertheless useful as an independent check that feed has a strong negative effect on life.

7.7 NASA Material-Aware AFT

NASA is handled separately because its feed units, material families, fixed speed, and wear scale differ from NUAA. The material-aware AFT model includes feed, depth, material, and initial wear:

Table 7.4: NASA material-aware AFT model summary.

Term	Coefficient	Multiplicative effect
Intercept	4.0880	59.6220
log feed reference	-0.3874	0.6788
log depth reference	-1.1356	0.3212
material: steel	-1.6069	0.2005
initial wear	-3.1201	0.0442

The NASA AFT fit has event MAE 2.5498, RMSE 3.1544, $R^2 = 0.9689$, $\sigma = 0.2459$, and negative log likelihood 43.6327. Because all NASA runs cross the threshold in this local label set, the model is not a censored survival problem in the same way as NUAA, but it provides strong evidence for material and initial-wear effects.

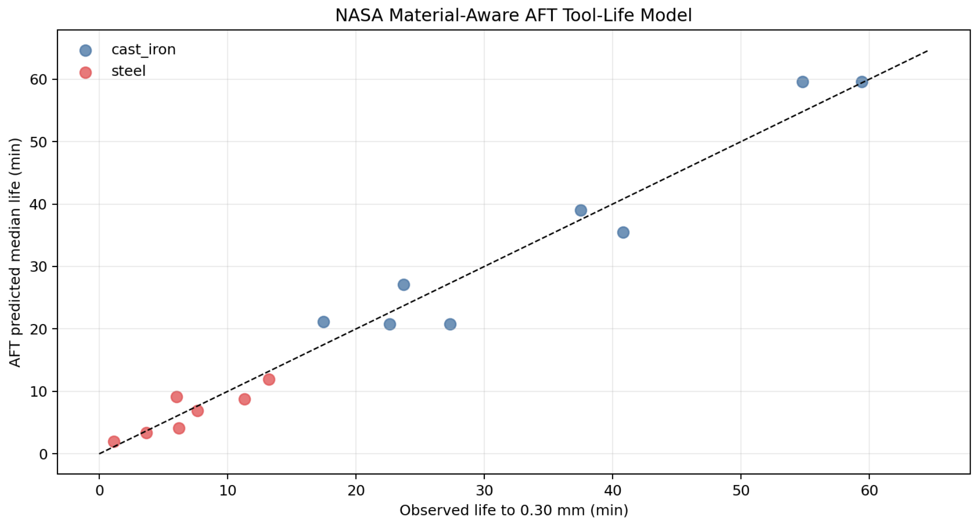


Figure 7.5: NASA material-aware AFT predictions. The high R^2 is promising but should be interpreted with the small case count and fixed-speed design in mind.

Chapter 8

Simulator and Digital-Twin Interpretation

8.1 From Equation to Simulator

The repository includes a simulator-oriented Phase 2 model and frontend artifacts. The core simulator behavior is driven by the piecewise law and the load-intensity function. The simulator should be understood as a digital twin prototype: it maps operating conditions to wear trajectories and life estimates, but its uncertainty bands and known failure modes should remain visible.

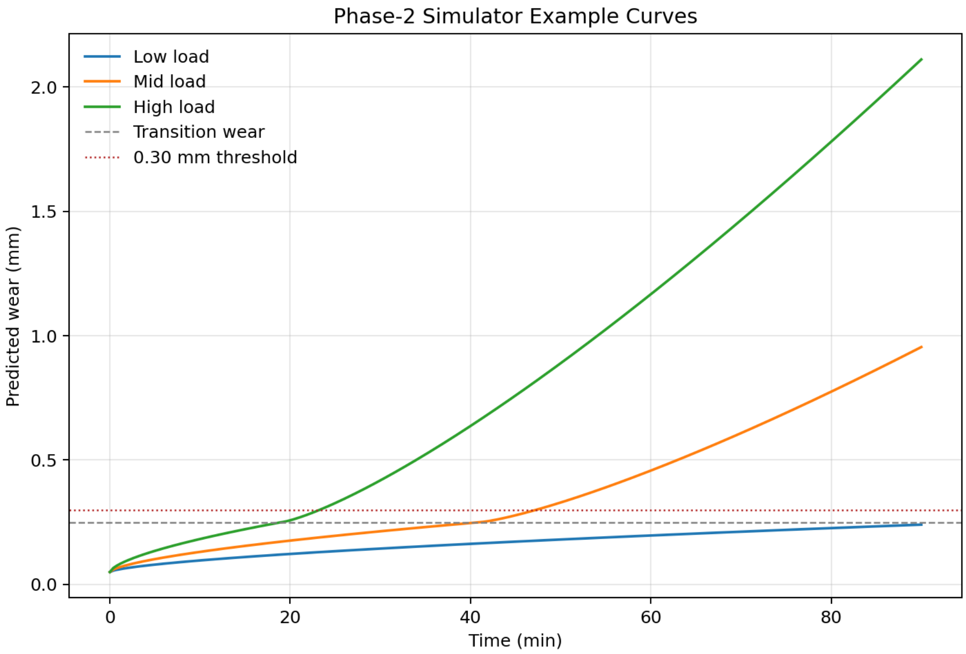


Figure 8.1: Example simulated wear curves. The piecewise law produces slow early growth followed by late acceleration.

8.2 Workload and Load-Effect Icons

The operational story can be summarized with three load states. The icons below are intentionally simple LaTeX/TikZ marks rather than external artwork, so the document remains fully editable and portable.

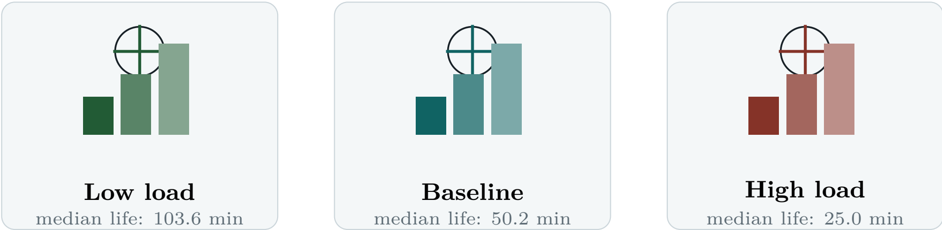


Figure 8.2: Load-effect icon comparison derived from bootstrap median life intervals. Higher feed/depth/speed compress predicted life.

8.3 Practical Decision Logic

The current digital-twin logic should be used for scenario ranking and research exploration:

1. Estimate the expected wear trajectory for a candidate load condition.
2. Track sensor health markers, especially vibration and AE modal features, for deviation from the expected trajectory.
3. Use survival-aware lifetime outputs when censoring or threshold crossing is the primary decision.
4. Treat high-speed exponents and short-window calibrations carefully, because they are less stable in the local data.

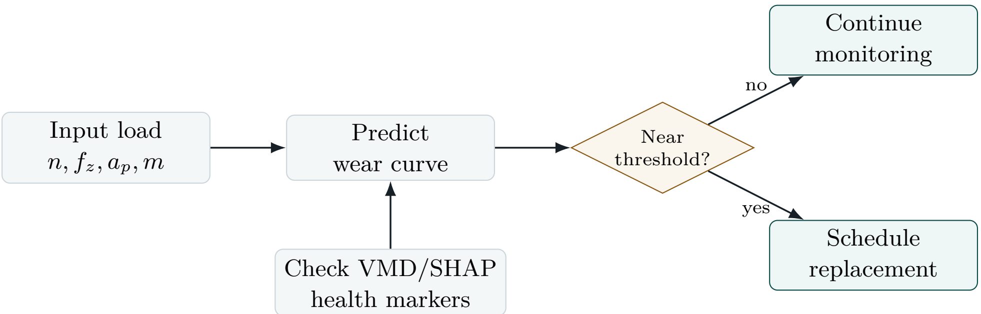


Figure 8.3: Decision logic implied by the research. The simulator should combine physics-based wear trajectory estimates with sensor health markers.

Chapter 9

External Research Context

9.1 Why Biology and Survival Modeling Matter

The lifetime rework borrowed ideas from survival analysis, including accelerated-failure-time modeling and censoring. This is common outside machining, especially in medicine and biology, where time-to-event data are often censored because patients leave a study or an event has not occurred by the observation cutoff. Tool-life data have the same structure: a tool may not reach the failure threshold before the experiment ends. Treating those runs as ordinary non-events wastes information and biases the model.

Deep survival models, Gaussian-process remaining-useful-life methods, physics-informed Gaussian processes, and physics-informed neural networks are useful pivots from adjacent domains. The repository does not blindly import these methods. Instead, it uses their modeling idea: preserve censoring, quantify uncertainty, and embed physical constraints where the data are too sparse to identify a purely black-box model.

9.2 Literature Positioning

Table 9.1: External literature themes and how they influence this project.

Theme	Representative source type	Influence on this research
PHM milling benchmarks	PHM competition and benchmark reviews	Motivates grouped validation, run-level splits, and caution about leakage across adjacent samples.
Public full-life milling datasets	2025 Scientific Data milling datasets	Identifies next ingestion targets for broader validation beyond the three local datasets.
Survival analysis	DeepSurv and accelerated-failure-time literature	Provides a principled vocabulary for event labels, right censoring, and life prediction under incomplete runs.
RUL uncertainty	PHM RUL uncertainty and Gaussian-process RUL work	Motivates prediction intervals and coefficient bootstrap rather than single deterministic life numbers.
Physics-informed learning	PINNs, physics-informed GP, and PI-KAF tool-wear monitoring	Supports the project choice to combine a mechanistic wear equation with statistical features and ML models.
Frequency decomposition	VMD and modal signal-processing literature	Supports modal features as interpretable signal evidence for wear progression.

Chapter 10

Limitations and Research Risk

10.1 Identified Limitations

1. The speed exponent is unstable in bootstrap and extracted life equations because the local speed range is narrow and correlated with other design variables.
2. NASA provides strong late-stage and sensor evidence, but its feed units, material labels, and fixed-speed design make direct pooling with NUAA unsafe.
3. PHM2010 is valuable for early-regime transfer, but the local use of PHM2010 has only three cutter groups.
4. Gradient Boosting has the best NUAA pointwise wear RMSE but worse event-life MAE after threshold inversion, so model selection must follow the target decision.
5. Per-run scalar calibration from short early windows overcorrects in NUAA holdout tests.
6. The simulator is calibrated to the local datasets and should not be treated as an industrial replacement policy until validated on newer full-life milling datasets.

10.2 Risk Matrix

Table 10.1: Research risk matrix.

Risk	Severity	Current evidence	Recommended response
Speed exponent over-interpretation	High	Wide bootstrap interval and large ML-inverted exponent	Report speed sensitivity as provisional and prioritize new experiments with wider speed variation.
Dataset pooling mismatch	High	Different units, sensors, and thresholds	Maintain dataset-specific roles and use transfer learning only after unit harmonization.
Calibration overcorrection	Medium	Calibrated RMSE worse than uncalibrated in NUAA	Replace scalar calibration with Bayesian or state-space updating.
Small event count	High	NUAA has 5 events and 4 censored records	Add full-life datasets and keep censoring explicit.
VMD feature overfitting	Medium	Many modal features relative to NASA sample count	Use grouped validation, feature stability, and sparse model variants.
Simulator misuse	Medium	Simulator curves look decisive despite uncertainty	Surface intervals, limits, and dataset provenance in the UI.

Chapter 11

Recommended Next Experiments

11.1 Controlled Data Expansion

The next research step is not to add more model classes first. It is to ingest broader full-life milling data under a strict provenance and validation plan. The 2025 QIT-CEMC and other open milling datasets listed in the research log are natural candidates. Each new dataset should be mapped into a common schema:

- tool identifier, material, coating, cutter geometry, and workpiece material;
- cutting speed or spindle speed, feed per tooth or feed rate, axial/radial depth;
- wear measurement type, threshold definition, and native time base;
- available sensor channels, sampling rate, and preprocessing;
- event/censoring indicator at the chosen threshold.

11.2 Modeling Roadmap

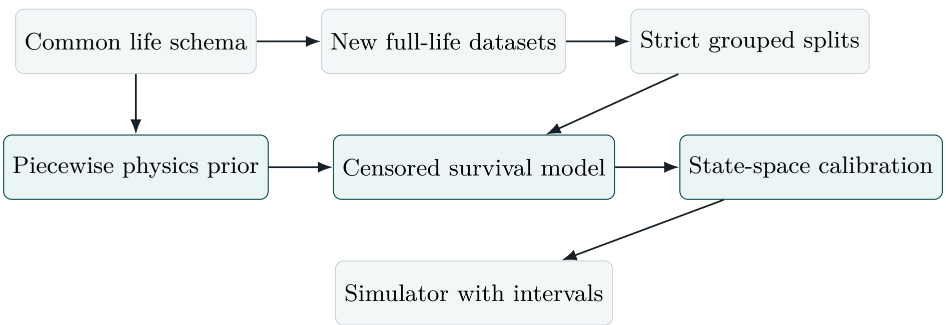


Figure 11.1: Recommended next modeling roadmap. The highest-value next work is structured dataset ingestion and survival-aware calibration, not simply adding more regressors.

11.3 Concrete Experiments

1. Refit the Phase 2 law after adding at least one full-life 2025 milling dataset.
2. Run leave-dataset-out validation to measure whether the piecewise exponents transfer.
3. Replace scalar calibration with a monotonic state-space update using sensor residuals.
4. Train survival models that combine physics-law covariates with VMD feature summaries.
5. Compare decision-level metrics: replacement lead time, missed-failure rate, false replacement rate, and expected unused tool life.

6. Add uncertainty display to the simulator frontend so users see median and interval curves together.

Chapter 12

Conclusion

The repository now contains a coherent research program for milling tool wear and tool-life forecasting. The strongest technical result is not a single machine-learning score; it is the combination of interpretable physics, signal evidence, grouped validation, threshold inversion, and censoring-aware life modeling. The piecewise wear law explains why early and late data should not be forced into one undifferentiated fit. The VMD and SHAP analyses show which sensor signatures carry wear information. The lifetime extraction pass shows that training a model and inverting it into life is possible, but also that the best wear regressor is not necessarily the best life predictor.

The practical interpretation is clear. Feed is the most reliable local load driver. Late-stage material behavior matters. Censoring must be modeled rather than ignored. Short-window scalar calibration is not yet dependable. The current digital twin is ready for research simulation, scenario ranking, and explainer visualizations, but it needs larger full-life validation before industrial deployment.

Appendix A

Generated Artifact Index

Table A.1: Key local artifacts used in this monograph.

Artifact	Path	Purpose
Phase 2 summary	`phase-2/outputs/phase2_model_summary.json`	Core piecewise-law coefficients and plots.
Extended report	`phase-2/outputs/extended_research/phase2_extended_research_report.md`	Bootstrap, intervals, calibration, residuals.
Lifetime report	`phase-2/outputs/lifetime_modeling/phase2_lifetime_model_report.md`	Trained-model inversion and AFT results.
Cross-dataset report	`results/cross_dataset_research/tool_health_failure_research_report.md`	Regression, classification, and failure marker results.
VMD summary	`results/vmd/vmd_summary.txt`	Modal feature extraction and correlation summary.
Interpretation summary	`results/interpretation/interpretation_summary.txt`	SHAP and ablation evidence.
Research log	`phase-2/research_log.md`	Chronological record of extension and lifetime-modeling work.
Sources log	`phase-2/sources.md`	Local and external source inventory.

Appendix B

Equation Summary

Table B.1: Main equations produced by the research.

Model	Equation
Condition intensity	$\phi = (n/1800)^{0.359967}(f_z/0.05)^{3.956013}(a_p/3.0)^{0.688073}$
Early wear	$VB(t) = 0.018580943\,\phi\,t^{0.64}$
NUAA censored AFT	$T_{\min} = 24.93(n/1800)^{-0.0187}(f_z/0.05)^{-1.7662}(a_p/3.0)^{-0.3768}$
Ridge-inverted life surface	$T_{\min} = 47.52(n/1800)^{-10.4924}(f_z/0.05)^{-4.6506}(a_p/3.0)^{-0.1372}$
Observed-life equation	$T_{\min} = 16.78(n/1800)^{-12.6655}(f_z/0.05)^{-4.1634}(a_p/3.0)^{-2.4111}$
NUAA Taylor-like observed lives	$T \approx 1.573\times10^{28}\, n^{-9.2806} f_z^{-4.6482} a_p^{-2.0471}$

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